

C4: Yingchuan parameter checkpoint lambda=0.4, beta=0.0015

Sapelo array 48482278; five prespecified exploratory seeds. This is one of eight sequential checkpoints. All five jobs are terminal before this report; no next beta was launched for its construction.

Evidence boundary: measured CIFAR-100-C severity-5 trajectories and finite intervention responses. Numerical failure is a failed computation, not behavioral deterioration. A finite F response does not identify a closed Jacobian, spectrum or phase transition.

Complete five-seed status

Seed	Parent blocks	Extension blocks	Forks verified	H2O analyses	Coverage / reason
202	15	30	4/4	2/2	complete
303	15	30	4/4	2/2	complete
404	15	30	4/4	2/2	complete
505	15	30	4/4	2/2	complete
606	15	30	4/4	2/2	complete

A component is reported only after its own completion and verification. Parent, extension, fork and H2O gaps remain explicit; the raw run and failed attempts remain on Sapelo.

Exploratory readout: visit-3 minus visit-1 same-domain CE is positive in 2/5 seeds (mean +0.014); classification error changes by -4.38 percentage points on average. At H1410, 7/10 correlated parent nodes have harmful closed acceptance and 8/10 have positive feedback interaction. The signs are mixed; these are descriptive, not a persistent-deterioration or stability claim.

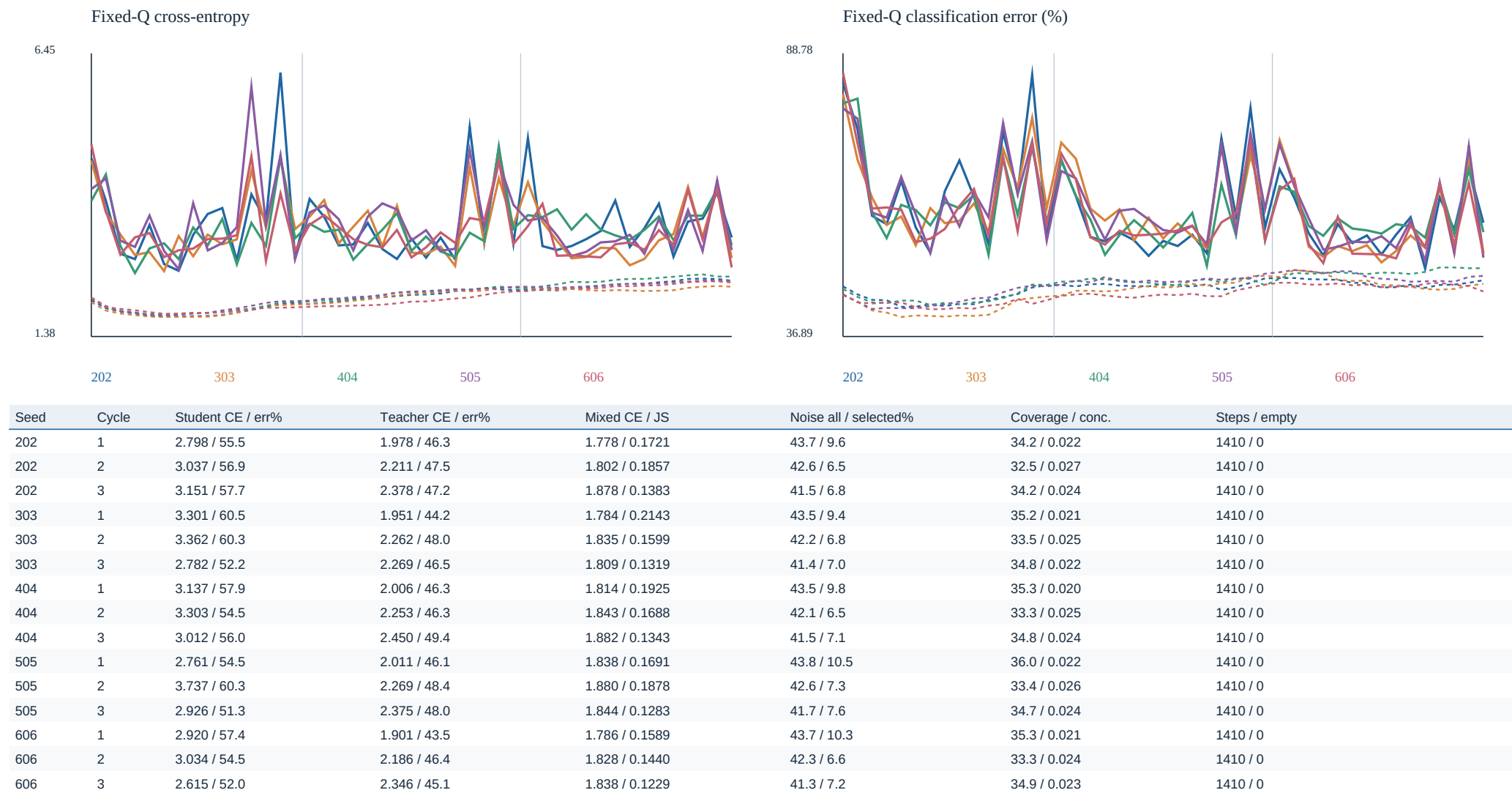
C4-1. Proxy versus labelled local update

Negative post-minus-pre CE means local improvement. Near zero means either absolute change is below 1e-4. Natural C4 executes updates without a separate accept/reject evaluator.

Seed	Probes	Both improve	Proxy down / local up	Proxy up / local down	Both worsen	Near zero
202	90	47	42	0	1	0
303	90	50	39	0	0	1
404	90	45	44	0	1	0
505	90	56	33	0	1	0
606	90	45	44	1	0	0

C4-2. Complete fixed-Q trajectories and pseudo-supervision

The fixed-Q panel uses the same 1,500 image-corruption evaluations at every block. Curves are unsmoothed. Dotted curves are teachers; solid curves are students. Missing/failed cells are absent, not assigned zero risk.



Cycle CSV retains teacher/student disagreement, mixed-target confidence, count-weighted pseudo-label statistics, selected-class concentration and elapsed block time. Selected noise is undefined when no examples were selected.

C4-3. Same-domain returns

Visit-3 minus visit-1 post-block student CE for each named corruption. Positive means worse on that domain; all domains and all terminal cells are retained. The companion block CSV includes pre/post CE and error for student and teacher at every visit.

Corruption	202	303	404	505	606
gaussian_noise	1.410	-0.583	-0.021	-0.540	-2.498
shot_noise	-0.888	0.077	-0.056	-0.480	0.532
impulse_noise	0.443	0.236	0.599	0.283	0.234
defocus_blur	0.443	0.137	0.419	0.321	0.416
glass_blur	-0.144	-0.012	0.667	-1.325	-0.299
motion_blur	0.528	0.416	0.293	0.405	0.447
zoom_blur	0.678	0.145	0.366	0.388	0.432
snow	-0.392	0.186	-0.262	-0.745	0.249
frost	-0.609	-0.872	0.054	-0.103	0.289
fog	-0.169	0.348	0.480	0.722	0.417
brightness	0.354	0.388	0.373	-0.194	0.280
contrast	-0.130	-0.760	-0.154	-0.264	-0.218
elastic_transform	0.165	0.457	0.166	0.078	0.422
pixelate	-2.612	-0.068	-0.048	-0.355	-0.047
jpeg_compression	0.242	-0.223	0.044	0.184	-0.107

C4-4. Per-seed parameter summary

Seed	Mean post CE / err%	Mean return dCE / dErr pp	Cycle-3 CE / err%	Coverage%	Updates	Block cost min
202	2.917 / 54.7	-0.046 / -6.53	2.907 / 51.8	33.6	4230	15.3
303	2.850 / 53.4	-0.009 / -5.16	2.801 / 51.0	34.5	4230	15.1
404	2.764 / 53.0	0.195 / -3.45	2.859 / 51.4	34.5	4230	15.3
505	2.929 / 54.5	-0.108 / -3.39	2.808 / 52.8	34.7	4230	15.6
606	2.782 / 53.0	0.037 / -3.38	2.795 / 51.6	34.5	4230	15.0

C4-5. Matched accept/reject and closed/replay forks

The first nonempty update after preselected blocks 5 and 15 is the focal candidate. Positive accept-minus-reject CE is worse. $A_{\text{closed}} = \text{accept_closed} - \text{reject_closed}$; $A_{\text{open}} = \text{accept_replay} - \text{reject_replay}$; $I_{\text{feedback}} = A_{\text{closed}} - A_{\text{open}}$. The H94 and H1410 forks share their initial state and schedule; their latest common scored checkpoint, H93, agrees exactly. H94 is therefore a correlated branch, not an independent replicate. The H1410 run does not separately score H94.

Seed	Parent	Immediate Q dCE / dErr pp	Immediate domain dCE / dErr pp
202	5	-0.042 / 0.07	-0.062 / -0.60
202	15	0.046 / 1.27	-0.094 / -0.47
303	5	-0.016 / 1.53	-0.042 / -1.00
303	15	-0.151 / -1.07	-0.370 / -3.07
404	5	-0.012 / -0.20	-0.008 / 0.53
404	15	0.079 / 0.47	-0.005 / 0.20
505	5	-0.146 / -0.93	-0.140 / -1.20
505	15	-0.055 / -0.47	-0.159 / -0.60
606	5	0.014 / 1.13	-0.017 / 1.33
606	15	0.068 / 0.40	0.206 / 1.00

H94 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_{closed}	A_{open}	I_{feedback}
202	5	2.751	2.751	2.657	2.756	-0.094	0.005	-0.099
202	15	5.100	5.100	4.230	4.528	-0.870	-0.573	-0.298
303	5	2.987	2.987	2.559	2.731	-0.428	-0.256	-0.172
303	15	3.980	3.980	3.731	3.782	-0.249	-0.198	-0.051
404	5	3.298	3.298	3.050	3.126	-0.248	-0.172	-0.076
404	15	3.503	3.503	3.592	3.235	0.088	-0.268	0.356
505	5	3.065	3.065	2.966	2.941	-0.098	-0.123	0.025
505	15	4.002	4.002	3.691	3.937	-0.311	-0.065	-0.246
606	5	3.150	3.150	2.837	2.905	-0.313	-0.245	-0.069
606	15	3.576	3.576	3.272	3.134	-0.304	-0.442	0.138

H1409 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_closed	A_open	I_feedback
202	5	3.499	3.499	3.409	3.319	-0.089	-0.180	0.091
202	15	2.964	2.964	3.037	2.794	0.073	-0.170	0.243
303	5	3.447	3.447	3.629	3.407	0.182	-0.040	0.222
303	15	3.201	3.201	3.362	2.937	0.162	-0.264	0.426
404	5	3.294	3.294	3.035	2.828	-0.259	-0.466	0.207
404	15	2.735	2.735	3.303	2.849	0.568	0.114	0.455
505	5	3.267	3.267	3.524	3.230	0.257	-0.036	0.293
505	15	3.069	3.069	3.737	2.966	0.668	-0.103	0.770
606	5	3.189	3.189	3.019	3.136	-0.170	-0.053	-0.117
606	15	2.732	2.732	3.034	2.834	0.302	0.102	0.201

H1410 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_closed	A_open	I_feedback
202	5	3.498	3.498	3.263	3.339	-0.235	-0.159	-0.076
202	15	2.997	2.997	3.138	2.836	0.141	-0.161	0.302
303	5	3.469	3.469	3.796	3.340	0.327	-0.129	0.456
303	15	3.201	3.201	3.361	2.933	0.161	-0.268	0.429
404	5	3.390	3.390	3.026	2.896	-0.364	-0.494	0.131
404	15	2.713	2.713	3.332	2.842	0.619	0.129	0.490
505	5	3.237	3.237	3.496	3.220	0.259	-0.017	0.276
505	15	3.165	3.165	3.761	2.981	0.596	-0.184	0.780
606	5	3.229	3.229	3.056	3.179	-0.174	-0.050	-0.124
606	15	2.769	2.769	3.086	2.841	0.317	0.072	0.245

C4-6. Diagnostic state, finite F and local identifiability

x=(student fixed-Q CE, mixed-target fixed-Q CE, student/teacher JS, mixed confidence). F is the mean H20 future x over three base tapes at each observed parent; it is a conditional response, not a closed transition law. The block CSV stores every observed x_t .

Seed	Parent	x0	F_H20(x0)	Tape SD	Finite	J status
202	5	3.376 / 1.781 / 0.281 / 0.657	2.766 / 1.770 / 0.207 / 0.655	0.155 / 0.001 / 0.012 / 0.000	132/132	not identified
202	15	2.798 / 1.778 / 0.172 / 0.639	3.481 / 1.776 / 0.256 / 0.639	0.491 / 0.000 / 0.019 / 0.000	132/132	not identified
303	5	2.888 / 1.776 / 0.221 / 0.666	2.772 / 1.767 / 0.213 / 0.666	0.108 / 0.001 / 0.017 / 0.000	132/132	not identified
303	15	3.301 / 1.784 / 0.214 / 0.650	4.044 / 1.781 / 0.323 / 0.650	0.261 / 0.002 / 0.030 / 0.000	132/132	not identified
404	5	2.955 / 1.806 / 0.227 / 0.661	2.745 / 1.800 / 0.198 / 0.660	0.166 / 0.001 / 0.017 / 0.000	132/132	not identified
404	15	3.137 / 1.814 / 0.192 / 0.643	3.286 / 1.811 / 0.279 / 0.642	0.058 / 0.000 / 0.022 / 0.000	132/132	not identified
505	5	3.543 / 1.826 / 0.284 / 0.666	2.733 / 1.817 / 0.206 / 0.666	0.055 / 0.001 / 0.002 / 0.000	132/132	not identified
505	15	2.761 / 1.838 / 0.169 / 0.645	3.784 / 1.838 / 0.286 / 0.644	0.197 / 0.001 / 0.038 / 0.000	132/132	not identified
606	5	3.230 / 1.829 / 0.238 / 0.664	2.621 / 1.817 / 0.194 / 0.663	0.145 / 0.001 / 0.025 / 0.000	132/132	not identified
606	15	2.920 / 1.786 / 0.159 / 0.647	3.572 / 1.785 / 0.294 / 0.647	0.432 / 0.001 / 0.024 / 0.000	132/132	not identified

Perturbation, conditioning and held-out diagnostics

Seed	Parent	Median G	Rank	Condition	Scale diff	Held-out median	Closure	Reason
202	5	10.5	4	4.6	1.21	33.03	falsified	not established
202	15	67.8	4	3.9	0.59	309.38	falsified	not established
303	5	25.7	4	2.8	0.90	67.76	falsified	not established
303	15	39.7	4	2.1	0.91	172.83	falsified	not established
404	5	21.4	4	5.7	1.00	82.48	falsified	not established
404	15	28.2	4	5.3	0.91	84.04	falsified	not established
505	5	15.4	4	4.2	1.51	48.62	falsified	not established
505	15	68.3	4	3.3	1.18	173.84	falsified	not established
606	5	23.4	4	5.6	0.67	55.70	falsified	not established
606	15	64.7	4	2.3	0.90	334.44	falsified	not established

Independent student/teacher perturbations, input rank/condition, two scales, held-out directions and momentum closure checks are retained in the compact analysis files. No eigenvalue or phase boundary is claimed without a closed and predictive state law.

C4-7. Bounded parameter-map reading

This checkpoint isolates one pair $\lambda=0.4$, $\beta=0.0015$ with five exploratory seeds. Seeds are reused across later pairs for paired comparison; they are not independent cross-parameter replications. Outcomes at this pair alone cannot locate a stability boundary. Numerical failures and improvement/null findings remain in the reported denominator.

Observed coverage: 5/5 complete 45-block trajectories, 10 verified H1410 parent nodes, 7 positive A_{closed} and 8 positive I_{feedback} . Parent nodes within one seed are correlated. A positive endpoint or interaction alone does not establish sustained recursive deterioration.

Coverage ledger and missingness

C4 item	Present evidence	Missing / limitation
C4-1	450 completed local probes	Unavailable after failed/incomplete trajectories
C4-2	5 full trajectories and cycle process records	Failed/partial cells not extrapolated
C4-3	75 same-domain return comparisons	Only complete first/third visits
C4-4	5 per-seed parameter summaries	One pair cannot characterize a sweep
C4-5	30 fork horizons	Unavailable parent nodes explicitly excluded
C4-6	10 H20 parent analyses	J/spectrum only if closure and held-out criteria pass
C4-7	One-pair status/sign description	No identified region or boundary

Provenance: Sapelo 48482278, compact evidence MANIFEST.json and AUDIT.json. Raw checkpoints, logits, step tapes and all failures remain under /scratch/yz54720/RSI2RSD/C4/sweep_yingchuan_20260923/results/I04_b00015/. The reserved confirmation panel was not read.